

--Query-1 :- Total Sales--

```
SELECT ROUND(SUM(sales), 2) AS total_sales  
FROM shopsphere_sales;
```

	total_sales numeric
1	2297200.86

--Query-2 :- Total Profit--

```
SELECT ROUND(SUM(profit), 2) AS total_profit  
FROM shopsphere_sales;
```

	total_profit numeric
1	286397.02

--Query-3 :- Total Orders (distinct orders, not row count
-- since one order has multiple line items)

```
SELECT COUNT(DISTINCT order_id) AS total_orders  
FROM shopsphere_sales;
```

	total_orders bigint
1	5009

--Query-4 :- Average Order Value (sales summed per order first, then averaged --
-- not just averaging every row's sales)--

```
WITH order_totals AS (  
  SELECT order_id, SUM(sales) AS order_value  
  FROM shopsphere_sales  
  GROUP BY order_id  
)  
SELECT ROUND(AVG(order_value), 2) AS avg_order_value  
FROM order_totals;
```

	avg_order_value numeric
1	458.61

--Query-5 :- Total Unique Customers--

```
SELECT COUNT(DISTINCT customer_id) AS total_unique_customers
FROM shopsphere_sales;
```

	total_unique_customers bigint 
1	793

--Query-6 :- Sales by Category--

```
SELECT category, ROUND(SUM(sales), 2) AS total_sales
FROM shopsphere_sales
GROUP BY category
ORDER BY total_sales DESC;
```

	category character varying (20) 	total_sales numeric 
1	Technology	836154.03
2	Furniture	741999.80
3	Office Supplies	719047.03

--Query-7 :- Profit by Category--

```
SELECT category, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY category
ORDER BY total_profit DESC;
```

	category character varying (20) 	total_profit numeric 
1	Technology	145454.95
2	Office Supplies	122490.80
3	Furniture	18451.27

--Query-8 :- Sales by Sub-Category--

```
SELECT sub_category, ROUND(SUM(sales), 2) AS total_sales
FROM shopsphere_sales
GROUP BY sub_category
```

ORDER BY total_sales DESC;

	sub_category character varying (20) 🔒	total_sales numeric 🔒
1	Phones	330007.05
2	Chairs	328449.10
3	Storage	223843.61
4	Tables	206965.53
5	Binders	203412.73
6	Machines	189238.63
7	Accessories	167380.32
8	Copiers	149528.03
9	Bookcases	114880.00
10	Appliances	107532.16
11	Furnishings	91705.16
12	Paper	78479.21
13	Supplies	46673.54
14	Art	27118.79
15	Envelopes	16476.40
16	Labels	12486.31
17	Fasteners	3024.28

--Query-9 :- Profit by Sub-Category--

```
SELECT sub_category, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY sub_category
```

ORDER BY total_profit DESC;

	sub_category character varying (20) 🔒	total_profit numeric 🔒
1	Copiers	55617.82
2	Phones	44515.73
3	Accessories	41936.64
4	Paper	34053.57
5	Binders	30221.76
6	Chairs	26590.17
7	Storage	21278.83
8	Appliances	18138.01
9	Furnishings	13059.14
10	Envelopes	6964.18
11	Art	6527.79
12	Labels	5546.25
13	Machines	3384.76
14	Fasteners	949.52
15	Supplies	-1189.10
16	Bookcases	-3472.56
17	Tables	-17725.48

--Query-10 :- Top 10 Products by Sales--

```
SELECT product_name, ROUND(SUM(sales), 2) AS total_sales
FROM shopsphere_sales
GROUP BY product_name
ORDER BY total_sales DESC
```

LIMIT 10;

	product_name character varying (200)	total_sales numeric
1	Canon imageCLASS 2200 Advanced Copier	61599.82
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.38
3	Cisco TelePresence System EX90 Videoconferencing Unit	22638.48
4	HON 5400 Series Task Chairs for Big and Tall	21870.58
5	GBC DocuBind TL300 Electric Binding System	19823.48
6	GBC Ibimaster 500 Manual ProClick Binding System	19024.50
7	Hewlett Packard LaserJet 3310 Copier	18839.69
8	HP Designjet T520 Inkjet Large Format Printer - 24" Color	18374.90
9	GBC DocuBind P400 Electric Binding System	17965.07
10	High Speed Automatic Electric Letter Opener	17030.31

--Query-11 :- Bottom 10 Products by Profit (biggest loss-makers) --

```
SELECT product_name, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY product_name
ORDER BY total_profit ASC
LIMIT 10;
```

	product_name character varying (200)	total_profit numeric
1	Cubify CubeX 3D Printer Double Head Print	-8879.97
2	Lexmark MX611dhe Monochrome Laser Printer	-4589.97
3	Cubify CubeX 3D Printer Triple Head Print	-3839.99
4	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	-2876.12
5	Bush Advantage Collection Racetrack Conference Table	-1934.40
6	GBC DocuBind P400 Electric Binding System	-1878.17
7	Cisco TelePresence System EX90 Videoconferencing Unit	-1811.08
8	Martin Yale Chadless Opener Electric Letter Opener	-1299.18
9	Balt Solid Wood Round Tables	-1201.06
10	BoxOffice By Design Rectangular and Half-Moon Meeting Room Tables	-1148.44

--Query-12 :- Most Sold Products (by Quantity) --

```
SELECT product_name, SUM(quantity) AS total_quantity_sold
```

```

FROM shoppisphere_sales
GROUP BY product_name
ORDER BY total_quantity_sold DESC
LIMIT 10;

```

	product_name character varying (200) 🔒	total_quantity_sold bigint 🔒
1	Staples	215
2	Staple envelope	170
3	Easy-staple paper	150
4	Staples in misc. colors	86
5	KI Adjustable-Height Table	74
6	Avery Non-Stick Binders	71
7	Storex Dura Pro Binders	71
8	GBC Premium Transparent Covers with Diagonal Lined Pattern	67
9	Situations Contoured Folding Chairs, 4/Set	64
10	Staple-based wall hangings	62

--Query-13 :- Top Customers by Sales --

```

SELECT customer_id, ROUND(SUM(sales), 2) AS total_sales
FROM shoppisphere_sales
GROUP BY customer_id
ORDER BY total_sales DESC
LIMIT 10;

```

	customer_id character varying (10) 🔒	total_sales numeric 🔒
1	SM-20320	25043.05
2	TC-20980	19052.22
3	RB-19360	15117.34
4	TA-21385	14595.62
5	AB-10105	14473.57
6	KL-16645	14175.23
7	SC-20095	14142.33
8	HL-15040	12873.30
9	SE-20110	12209.44
10	CC-12370	12129.07

--Query-14 :- Top Customers by Profit --

```
SELECT customer_id, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY customer_id
ORDER BY total_profit DESC
LIMIT 10;
```

	customer_id character varying (10) 🔒	total_profit numeric 🔒
1	TC-20980	8981.32
2	RB-19360	6976.10
3	SC-20095	5757.41
4	HL-15040	5622.43
5	AB-10105	5444.81
6	TA-21385	4703.79
7	CM-12385	3899.89
8	KD-16495	3038.63
9	AR-10540	2884.62
10	DR-12940	2869.08

--Query-15 :- Number of Orders per Customer (distinct orders, not row count --
--same reasoning as before, since each order can have multiple product rows) --

```
SELECT customer_id, COUNT(DISTINCT order_id) AS num_orders
FROM shopsphere_sales
GROUP BY customer_id
ORDER BY num_orders DESC;
```

	customer_id character varying (10) 🔒	num_orders bigint 🔒
1	KB-16315	9
2	LC-16930	8
3	EB-13930	5
4	EH-13765	12
5	EB-13975	4
6	MM-18055	8
7	CS-12250	12
8	TP-21565	4
9	JS-15880	5
10	ML-17755	7

--Query-16 :- Average Spending per Customer --

```
SELECT customer_id, ROUND(AVG(sales), 2) AS avg_spending_per_line
FROM shopsphere_sales
GROUP BY customer_id
ORDER BY avg_spending_per_line DESC;
```

	customer_id character varying (10) 🔒	avg_spending_per_line numeric 🔒
1	JH-15430	44.18
2	MY-17380	173.41
3	JL-15850	247.75
4	JS-15595	177.21
5	SC-20845	70.16
6	PP-18955	213.31
7	JD-15790	307.48
8	FH-14275	202.68
9	MY-18295	197.75
10	IM-15055	718.16

--Query-17 :- Customer Distribution by Segment --

```

SELECT segment, COUNT(DISTINCT customer_id) AS num_customers
FROM shopsphere_sales
GROUP BY segment
ORDER BY num_customers DESC;

```

	customer_id character varying (10) 🔒	total_profit numeric 🔒	profit_rank bigint 🔒
1	TC-20980	8981.32	1
2	RB-19360	6976.10	2
3	SC-20095	5757.41	3
4	HL-15040	5622.43	4
5	AB-10105	5444.81	5
6	TA-21385	4703.79	6
7	CM-12385	3899.89	7
8	KD-16495	3038.63	8
9	AR-10540	2884.62	9
10	DR-12940	2869.08	10

--Query-18 :- Highest Profit Customers (same as #14, but showing top 10 explicitly with--
--ranking) --

```

SELECT
    customer_id,
    ROUND(SUM(profit), 2) AS total_profit,
    RANK() OVER (ORDER BY SUM(profit) DESC) AS profit_rank
FROM shopsphere_sales
GROUP BY customer_id
ORDER BY total_profit DESC

```

LIMIT 10;

	customer_id character varying (10) 🔒	total_profit numeric 🔒	profit_rank bigint 🔒
1	TC-20980	8981.32	1
2	RB-19360	6976.10	2
3	SC-20095	5757.41	3
4	HL-15040	5622.43	4
5	AB-10105	5444.81	5
6	TA-21385	4703.79	6
7	CM-12385	3899.89	7
8	KD-16495	3038.63	8
9	AR-10540	2884.62	9
10	DR-12940	2869.08	10

--Query-19 :- Sales by Region --

```
SELECT region, ROUND(SUM(sales), 2) AS total_sales
FROM shopsphere_sales
GROUP BY region
ORDER BY total_sales DESC;
```

	region character varying (10) 🔒	total_sales numeric 🔒
1	West	725457.82
2	East	678781.24
3	Central	501239.89
4	South	391721.91

--Query-20 :- Profit by Region --

```
SELECT region, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY region
```

ORDER BY total_profit DESC;

	region character varying (10) 🔒	total_profit numeric 🔒
1	West	108418.45
2	East	91522.78
3	South	46749.43
4	Central	39706.36

--Query-21 :- Sales by State --

```
SELECT state, ROUND(SUM(sales), 2) AS total_sales
FROM shopsphere_sales
GROUP BY state
ORDER BY total_sales DESC;
```

	state character varying (30) 🔒	total_sales numeric 🔒
1	California	457687.63
2	New York	310876.27
3	Texas	170188.05
4	Washington	138641.27
5	Pennsylvania	116511.91
6	Florida	89473.71
7	Illinois	80166.10
8	Ohio	78258.14
9	Michigan	76269.61
10	Virginia	70636.72
11	North Carolina	55603.16
12	Indiana	53555.36
13	Georgia	49095.84
14	Kentucky	36591.75
15	New Jersey	35764.31
16	Arizona	35282.00
17	Wisconsin	32114.61
18	Colorado	32108.12
19	Tennessee	30661.87
20	Minnesota	29863.15
21	Massachusetts	28634.43
22	Delaware	27451.07
23	Maryland	23705.52
24	Rhode Island	22627.96
25	Missouri	22205.15
26	Oklahoma	19683.39

27	Alabama	19510.64
28	Oregon	17431.15
29	Nevada	16729.10
30	Connecticut	13384.36
31	Arkansas	11678.13
32	Utah	11220.06
33	Mississippi	10771.34
34	Louisiana	9217.03
35	Vermont	8929.37
36	South Carolina	8481.71
37	Nebraska	7464.93
38	New Hampshire	7292.52
39	Montana	5589.35
40	New Mexico	4783.52
41	Iowa	4579.76
42	Idaho	4382.49
43	Kansas	2914.31
44	District of Columbia	2865.02
45	Wyoming	1603.14
46	South Dakota	1315.56
47	Maine	1270.53
48	West Virginia	1209.82
49	North Dakota	919.91

--Query-22 :- Profit by State --

```

SELECT state, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY state
ORDER BY total_profit DESC;

```

	state character varying (30) 🔒	total_profit numeric 🔒
1	California	76381.39
2	New York	74038.55
3	Washington	33402.65
4	Michigan	24463.19
5	Virginia	18597.95
6	Indiana	18382.94
7	Georgia	16250.04
8	Kentucky	11199.70
9	Minnesota	10823.19
10	Delaware	9977.37
11	New Jersey	9772.91
12	Wisconsin	8401.80
13	Rhode Island	7285.63
14	Maryland	7031.18
15	Massachusetts	6785.50
16	Missouri	6436.21
17	Alabama	5786.83
18	Oklahoma	4853.96
19	Arkansas	4008.69
20	Connecticut	3511.49
21	Nevada	3316.77
22	Mississippi	3172.98
23	Utah	2546.53
24	Vermont	2244.98
25	Louisiana	2196.10
26	Nebraska	2037.09

27	Montana	1833.33
28	South Carolina	1769.06
29	New Hampshire	1706.50
30	Iowa	1183.81
31	New Mexico	1157.12
32	District of Columbia	1059.59
33	Kansas	836.44
34	Idaho	826.72
35	Maine	454.49
36	South Dakota	394.83
37	North Dakota	230.15
38	West Virginia	185.92
39	Wyoming	100.20
40	Oregon	-1190.47
41	Florida	-3399.30
42	Arizona	-3427.92
43	Tennessee	-5341.69
44	Colorado	-6527.86
45	North Carolina	-7490.91
46	Illinois	-12607.89
47	Pennsylvania	-15559.96
48	Ohio	-16971.38
49	Texas	-25729.36

--Query-23 :- Top 10 States by Profit --

```

SELECT state, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY state
ORDER BY total_profit DESC
LIMIT 10;

```

	state character varying (30) 🔒	total_profit numeric 🔒
1	California	76381.39
2	New York	74038.55
3	Washington	33402.65
4	Michigan	24463.19
5	Virginia	18597.95
6	Indiana	18382.94
7	Georgia	16250.04
8	Kentucky	11199.70
9	Minnesota	10823.19
10	Delaware	9977.37

--Query-24 :- Loss-Making States --

```
SELECT state, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY state
HAVING SUM(profit) < 0
ORDER BY total_profit ASC;
```

	state character varying (30) 🔒	total_profit numeric 🔒
1	Texas	-25729.36
2	Ohio	-16971.38
3	Pennsylvania	-15559.96
4	Illinois	-12607.89
5	North Carolina	-7490.91
6	Colorado	-6527.86
7	Tennessee	-5341.69
8	Arizona	-3427.92
9	Florida	-3399.30
10	Oregon	-1190.47

--Query-25 :- Monthly Sales Trend --

```
SELECT TO_CHAR(order_date, 'YYYY-MM') AS month,  
       ROUND(SUM(sales), 2) AS total_sales  
FROM shopsphere_sales  
GROUP BY TO_CHAR(order_date, 'YYYY-MM')  
ORDER BY month;
```

	month text	total_sales numeric			
1	2014-01	14236.90			
2	2014-02	4519.89			
3	2014-03	55691.01			
4	2014-04	28295.35			
5	2014-05	23648.29	27	2016-03	51715.88
6	2014-06	34595.13	28	2016-04	38750.04
7	2014-07	33946.39	29	2016-05	56987.73
8	2014-08	27909.47	30	2016-06	40344.53
9	2014-09	81777.35	31	2016-07	39261.96
10	2014-10	31453.39	32	2016-08	31115.37
11	2014-11	78628.72	33	2016-09	73410.02
12	2014-12	69545.62	34	2016-10	59687.75
13	2015-01	18174.08	35	2016-11	79411.97
14	2015-02	11951.41	36	2016-12	96999.04
15	2015-03	38726.25	37	2017-01	43971.37
16	2015-04	34195.21	38	2017-02	20301.13
17	2015-05	30131.69	39	2017-03	58872.35
18	2015-06	24797.29	40	2017-04	36521.54
19	2015-07	28765.33	41	2017-05	44261.11
20	2015-08	36898.33	42	2017-06	52981.73
21	2015-09	64595.92	43	2017-07	45264.42
22	2015-10	31404.92	44	2017-08	63120.89
23	2015-11	75972.56	45	2017-09	87866.65
24	2015-12	74919.52	46	2017-10	77776.92
25	2016-01	18542.49	47	2017-11	118447.83
26	2016-02	22978.82	48	2017-12	83829.32

--Query-26 :- Monthly Profit Trend --

```

SELECT TO_CHAR(order_date, 'YYYY-MM') AS month,
       ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales

```

GROUP BY TO_CHAR(order_date, 'YYYY-MM')
ORDER BY month;

	month text	total_profit numeric
1	2014-01	2450.19
2	2014-02	862.31
3	2014-03	498.73
4	2014-04	3488.84
5	2014-05	2738.71
6	2014-06	4976.52
7	2014-07	-841.48
8	2014-08	5318.11
9	2014-09	8328.10
10	2014-10	3448.26
11	2014-11	9292.13
12	2014-12	8983.57
13	2015-01	-3281.01
14	2015-02	2813.85
15	2015-03	9732.10
16	2015-04	4187.50
17	2015-05	4667.87
18	2015-06	3335.56
19	2015-07	3288.65
20	2015-08	5355.81
21	2015-09	8209.16
22	2015-10	2817.37
23	2015-11	12474.79
24	2015-12	8016.97
25	2016-01	2824.82
26	2016-02	5004.58
27	2016-03	3611.97
28	2016-04	2977.81
29	2016-05	8662.15
30	2016-06	4750.38
31	2016-07	4432.88
32	2016-08	2062.07
33	2016-09	9328.66
34	2016-10	16243.14
35	2016-11	4011.41
36	2016-12	17885.31
37	2017-01	7140.44
38	2017-02	1613.87
39	2017-03	14751.89
40	2017-04	933.29
41	2017-05	6342.58
42	2017-06	8223.34
43	2017-07	6952.62
44	2017-08	9040.96
45	2017-09	10991.56
46	2017-10	9275.28
47	2017-11	9690.10
48	2017-12	8483.35

--Query-27 :- Yearly Sales --

```

SELECT EXTRACT(YEAR FROM order_date) AS year,
       ROUND(SUM(sales), 2) AS total_sales
FROM shopsphere_sales
GROUP BY EXTRACT(YEAR FROM order_date)
ORDER BY year;

```

	year numeric 	total_sales numeric 
1	2014	484247.50
2	2015	470532.51
3	2016	609205.60
4	2017	733215.26

--Query-28 :- Quarterly Sales --

```

SELECT EXTRACT(YEAR FROM order_date) AS year,
       EXTRACT(QUARTER FROM order_date) AS quarter,
       ROUND(SUM(sales), 2) AS total_sales
FROM shopsphere_sales
GROUP BY EXTRACT(YEAR FROM order_date), EXTRACT(QUARTER FROM
order_date)

```

ORDER BY year, quarter;

	year numeric	quarter numeric	total_sales numeric
1	2014	1	74447.80
2	2014	2	86538.76
3	2014	3	143633.21
4	2014	4	179627.73
5	2015	1	68851.74
6	2015	2	89124.19
7	2015	3	130259.58
8	2015	4	182297.01
9	2016	1	93237.18
10	2016	2	136082.30
11	2016	3	143787.36
12	2016	4	236098.75
13	2017	1	123144.86
14	2017	2	133764.37
15	2017	3	196251.96
16	2017	4	280054.07

--Query-29 :- Best Sales Month --

```
SELECT TO_CHAR(order_date, 'YYYY-MM') AS month,  
       ROUND(SUM(sales), 2) AS total_sales  
FROM shopsphere_sales  
GROUP BY TO_CHAR(order_date, 'YYYY-MM')  
ORDER BY total_sales DESC  
LIMIT 1;
```

	month text	total_sales numeric
1	2017-11	118447.83

--Query-30 :- Best Profit Month --

```
SELECT TO_CHAR(order_date, 'YYYY-MM') AS month,  
       ROUND(SUM(profit), 2) AS total_profit
```

```

FROM shopsphere_sales
GROUP BY TO_CHAR(order_date, 'YYYY-MM')
ORDER BY total_profit DESC
LIMIT 1;

```

	month text	total_profit numeric
1	2016-12	17885.31

--Query-31 :- Average Discount by Category --

```

SELECT category, ROUND(AVG(discount), 4) AS avg_discount
FROM shopsphere_sales
GROUP BY category
ORDER BY avg_discount DESC;

```

	category character varying (20)	avg_discount numeric
1	Furniture	0.1739
2	Office Supplies	0.1573
3	Technology	0.1323

--Query-32 :- Profit by Discount Level --

```

SELECT
CASE
    WHEN discount = 0 THEN '0% (No Discount)'
    WHEN discount <= 0.20 THEN '1-20%'
    WHEN discount <= 0.40 THEN '21-40%'
    WHEN discount <= 0.60 THEN '41-60%'
    ELSE '60%+'
END AS discount_level,
ROUND(SUM(profit), 2) AS total_profit,
COUNT(*) AS num_rows
FROM shopsphere_sales
GROUP BY discount_level

```

ORDER BY discount_level;

	discount_level text	total_profit numeric	num_rows bigint
1	0% (No Discount)	320987.60	4798
2	1-20%	100785.47	3803
3	21-40%	-35817.47	460
4	41-60%	-28944.19	215
5	60%+	-70614.40	718

--Query-33 :- Average Shipping Days --

```
SELECT ROUND(AVG(ship_date - order_date), 2) AS avg_shipping_days
FROM shopsphere_sales;
```

	avg_shipping_days numeric
1	3.96

--Query-34 :- Shipping Days by Ship Mode --

```
SELECT ship_mode,
       ROUND(AVG(ship_date - order_date), 2) AS avg_shipping_days
FROM shopsphere_sales
GROUP BY ship_mode
ORDER BY avg_shipping_days;
```

	ship_mode character varying (20)	avg_shipping_days numeric
1	Same Day	0.04
2	First Class	2.18
3	Second Class	3.24
4	Standard Class	5.01

--Query-35 :- Profit by Ship Mode--

```
SELECT ship_mode, ROUND(SUM(profit), 2) AS total_profit
FROM shopsphere_sales
GROUP BY ship_mode
```

ORDER BY total_profit DESC;

	ship_mode character varying (20) 🔒	total_profit numeric 🔒
1	Standard Class	164088.79
2	Second Class	57446.64
3	First Class	48969.84
4	Same Day	15891.76

--Query-36 :- Discount Impact on Profit--

-- Overall correlation: negative value = higher discount tends to mean lower profit
 SELECT ROUND(CORR(discount, profit)::numeric, 4) AS discount_profit_correlation
 FROM shopsphere_sales;

-- Practical view: profit margin % by discount band (clearer story than raw totals)

```
SELECT
  CASE
    WHEN discount = 0 THEN '0% (No Discount)'
    WHEN discount <= 0.20 THEN '1-20%'
    WHEN discount <= 0.40 THEN '21-40%'
    WHEN discount <= 0.60 THEN '41-60%'
    ELSE '60%+'
  END AS discount_level,
  ROUND(AVG(profit / NULLIF(sales, 0)) * 100, 2) AS avg_profit_margin_pct
FROM shopsphere_sales
GROUP BY discount_level
ORDER BY discount_level;
```

	discount_level text 🔒	avg_profit_margin_pct numeric 🔒
1	0% (No Discount)	34.02
2	1-20%	17.44
3	21-40%	-16.69
4	41-60%	-63.41
5	60%+	-122.52